

DEV-594: Document Redis monitoring and alerting best practices

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Task: DEV-587-Redis Security Hardening

1. Executive Summary

This document defines the metrics Glynac should monitor on every Redis deployment and the alert thresholds tied to them. Metrics were collected live from a Redis 8.10.1 reference instance across INFO sections (memory, stats, clients, replication, persistence, keyspace, CPU), plus slowlog data captured from real commands. Two configuration gaps were found on the reference instance – `maxmemory: 0` (no memory ceiling) and `mem_fragmentation_ratio: 18.26` (high fragmentation) – both of which are alert-worthy conditions. Modern Redis also enables positive security defaults: `DEBUG` is blocked by default, preventing easy DoS via `DEBUG SLEEP`. The document closes with a monitoring stack recommendation and a Prometheus-compatible metric list for Glynac's observability platform.

2. Metric Categories

Redis exposes metrics through:

- **INFO** – snapshot of all internal counters (grouped by section)
- **SLOWLOG** – commands exceeding a time threshold
- **LATENCY** – latency spikes by event type
- **CLIENT LIST** – per-connection info
- **--stat mode** – rolling stats (CLI convenience)

Recommended stack for Glynac: **redis_exporter** → **Prometheus** → **Grafana** with **Alertmanager** routing. Alternatives: Datadog Redis integration, CloudWatch (for ElastiCache), New Relic.

3. Baseline Metrics (Reference Instance)

Captured live from `redis-cli INFO`:

Memory

```
used_memory: 1.45M
used_memory_rss: 26.50M
used_memory_peak: 1.45M
maxmemory: 0
maxmemory_policy: noeviction
mem_fragmentation_ratio: 18.26
```

Clients

```
connected_clients: 1
maxclients: 10000
blocked_clients: 0
```

Stats

```
total_connections_received: 7
total_commands_processed: 48
instantaneous_ops_per_sec: 10
rejected_connections: 0
keyspace_hits: 5
keyspace_misses: 2
expired_keys: 0
evicted_keys: 0
```

Replication

```
role: master
connected_slaves: 0
master_repl_offset: 202
```

Persistence

```
rdb_last_bgsave_status: ok
rdb_changes_since_last_save: 5
aof_enabled: 1
aof_last_write_status: ok
aof_last_bgrewrite_status: ok
aof_current_size: 432
```

Keyspace

```
db0: keys=8, expires=0, avg_ttl=0
```

ACL security events

```
acl_access_denied_auth: 0
acl_access_denied_cmd: 0
acl_access_denied_key: 0
```

4. Slowlog Test Results

Slowlog enabled at threshold 10000 μ s (10 ms). Temporarily lowered to 100 μ s to catch everything:

Duration (μ s)	Command
12,851	CONFIG SET slowlog-log-slower-than 100
3,866	SET slow:test1 abc
1,647	LPUSH slow:list a b c
799	LRANGE slow:list 0 -1

Findings:

- DEBUG SLEEP was **denied** by default – modern Redis blocks DEBUG unless explicitly enabled. This is a strong positive default.
- Even trivial commands logged at 100 μ s threshold – proving the slowlog mechanism works.
- In production, keep threshold at **10 ms** and alert if slowlog length grows faster than baseline.

5. Key Metrics to Monitor

#	Metric	Source	Why it matters
1	used_memory	INFO memory	Core capacity signal
2	maxmemory	INFO memory	Must be set (>0) to prevent OOM
3	mem_fragmentation_ratio	INFO memory	>1.5 → investigate; >3 → restart-worthy
4	evicted_keys	INFO stats	Eviction means pressure – size up or rethink TTLs
5	keyspace_hits / keyspace_misses	INFO stats	Hit rate < 80% → cache isn't effective
6	connected_clients	INFO clients	Approaching maxclients → connection starvation
7	rejected_connections	INFO stats	Any increase = clients being turned away
8	blocked_clients	INFO clients	Sustained > 0 → stuck consumer or slow client
9	instantaneous_ops_per_sec	INFO stats	Traffic baseline for anomaly detection
10	total_commands_processed	INFO stats	Throughput trending
11	rdb_last_bgsave_status	INFO persistence	err = backups failing
12	aof_last_write_status	INFO persistence	err = writes not persisting
13	aof_last_bgrewrite_status	INFO persistence	err = AOF rewrite failing
14	rdb_changes_since_l	INFO persistence	Growing =

	ast_save		snapshots not keeping up
15	connected_slaves	INFO replication	Should never drop below expected count
16	master_repl_offset vs replica offset	INFO replication	Lag = replication delay
17	acl_access_denied_*	INFO stats	Security signal – spikes = attack or misconfig
18	slowlog_commands_count	INFO stats	Growth = performance degradation
19	latest_fork_usec	INFO stats	High = COW cost from large dataset + BGSAVE
20	expired_keys	INFO stats	High = TTL churn – review key design

6. Alert Thresholds (Recommended)

Alert	Condition	Severity
No maxmemory set	maxmemory == 0 on production	Critical
Memory high	used_memory / maxmemory > 85% for 5 min	Warning
Memory critical	used_memory / maxmemory > 95% for 1 min	Critical
High fragmentation	mem_fragmentation_ratio > 1.5 for 15 min	Warning
Critical fragmentation	mem_fragmentation_ratio > 3	Critical
Evictions	evicted_keys > 0 (rate >0/s)	Warning
Low hit rate	hits / (hits + misses) < 80% for 10 min	Warning
Connection pressure	connected_clients / maxclients > 80%	Warning
Rejected connections	rejected_connections > 0 (any increase)	Critical
Blocked clients	blocked_clients > 0 for 5 min	Warning
RDB failure	rdb_last_bgsave_status == err	Critical
AOF write failure	aof_last_write_status == err	Critical
AOF rewrite failure	aof_last_bgrewrite_status == err	Critical
Snapshot lag	rdb_changes_since_last_save > 10000	Warning
Replica missing	connected_slaves < expected	Critical
Replication lag	master_repl_offset - replica_offset > 10MB	Warning

ACL denial spike	<code>acl_access_denied_* rate > 10/min</code>	Warning
ACL auth failures	<code>acl_access_denied_auth > 0</code>	Critical
Slowlog growth	<code>slowlog_commands_count grows > 10/min</code>	Warning
Fork latency	<code>latest_fork_usec > 1,000,000 (1s)</code>	Warning

Adjust numeric thresholds to Glynac's actual baseline after 2 weeks of observation.

7. Slowlog Configuration (Recommended)

```
# Log commands slower than 10 milliseconds
slowlog-log-slower-than 10000

# Keep last 512 entries
slowlog-max-len 512
```

Review cadence: Weekly via SLOWLOG GET 100. Investigate patterns – many KEYS, SORT, SMEMBERS on large sets, or HGETALL on large hashes.

8. Latency Monitoring

Redis tracks latency spikes per event type via the LATENCY command:

```
CONFIG SET latency-monitor-threshold 100 # 100 ms
LATENCY LATEST
LATENCY HISTORY <event>
LATENCY RESET
```

Key event types: command, fork, expire-cycle, eviction-del, aof-fsync-always, aof-write.

Alert if LATENCY LATEST shows any event > **100 ms** in the last 5 minutes.

9. Dashboard Recommendations (Grafana)

A single Redis dashboard should include:

Panel	Metric
Ops/sec	instantaneous_ops_per_sec
Memory usage	used_memory vs maxmemory
Hit rate	hits / (hits + misses)
Connected clients	connected_clients vs maxclients
Evictions/sec	rate(evicted_keys[1m])
Rejected conns/sec	rate(rejected_connections[1m])
Replication lag	master_repl_offset - slave_repl_offset
RDB/AOF status	Gauge (0=fail, 1=ok)
Slowlog count	slowlog_commands_count
ACL denials	All acl_access_denied_* counters
Keyspace size	db0:keys

10. Alert Routing

Severity	Route	Response time
Critical	PagerDuty / on-call page	Immediate
Warning	Slack #ops-alerts	Within business hours
Info	Slack #redis-metrics	Digest daily

11. Findings on Reference Instance

Finding	Severity	Action
maxmemory: 0	High	Set a hard ceiling before production
mem_fragmentation_ratio: 18.26	Medium	Expected on idle instance; monitor over time
connected_slaves: 0	Info	Expected – standalone lab; production needs replicas
aof_last_bgrewrite_status: ok	✓ Good	No action
rdb_last_bgsave_status: ok	✓ Good	No action
acl_access_denied_*: 0	✓ Good	No action
DEBUG blocked by default	✓ Good	Modern Redis defaults are safe
Slowlog configured (10ms)	✓ Good	Review weekly

12. Summary

Area	Outcome
Metrics collected	Memory, stats, clients, replication, persistence, keyspace, CPU, slowlog
Key findings	maxmemory=0; high fragmentation; DEBUG blocked by default
Slowlog test	4 entries captured, mechanism proven
Alerts defined	20 metrics with thresholds
Stack recommended	redis_exporter → Prometheus → Grafana → Alertmanager
Slowlog config	10ms threshold, 512 entries

13. Recommendations for Glynac

1. **Set maxmemory on every production instance** – the reference instance has none.
2. **Deploy redis_exporter + Prometheus + Grafana** for all Redis nodes.
3. **Alert on the 20 metrics defined in Section 6** – start with the Critical ones.
4. **Enable slowlog at 10ms** and review weekly.
5. **Monitor acl_access_denied_*** – spikes indicate attack or misconfig.
6. **Track hit rate** – below 80% means the cache isn't earning its keep.
7. **Set up alert routing** – Critical to pager, Warning to Slack.
8. **Review thresholds after 2 weeks** – tune against real baseline.

All metrics and slowlog outputs reproduced verbatim from a live Redis 8.10.1 reference instance.

END OF REPORT